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Multimodal Deep Learning for Oil Price Forecasting Using Economic Indicators

Siham AKIL[a], Sara SEKKATE[b], Abdellah ADIB[a,*]

[a]*Team: Data Science & Artificial Intelligence,Laboratory of Mathematics, Computer Science & Applications (LMCSA), Faculty of Sciences and Technologies Mohammedia, BP 146, Mohammedia, 28806, Morocco*

[b] *Higher National School of Arts and Crafts of Casablanca, 150 Bd du Nil, Casablanca 20670, Morocco*

Abstract

In the evolving landscape of global energy, accurately forecasting oil prices plays a pivotal role in strategizing the transition to green energy. Traditional forecasting methods, though widely used, often fall short in capturing the multifaceted influences on oil price dynamics. This research introduces a multimodal deep learning approach that incorporates both time series oil price data and key economic indicators to enhance forecasting accuracy. By employing a combination of Long Short-Term Memory (LSTM) and Temporal Convolutional Networks (TCN), the model adeptly captures temporal patterns and economic influences. The findings not only demonstrate the model's superior performance over traditional methods but also underscore the profound impact of specific economic indicators on oil prices.

Keywords: Multimodal Deep Learning, Oil price prediction, Economic Indicators, Time Series Analysis, Renewable Energy.

1. Introduction

The world's energy landscape is undergoing a profound transformation as the transition to renewable energy sources gains momentum [1]. As concerns about climate change and sustainability intensify, there's a global push towards green energy sources [2]. However, oil—often termed the "black gold"—continues to play a pivotal role in the global energy market, influencing global economy and geopolitics [3]. Its price fluctuations have far-reaching impacts, affecting everything from macroeconomic policies to household budgets [4].

Accurate oil price forecasting is crucial [5]. While traditional econometric models have provided insights, they sometimes struggle with the dynamic and multifaceted determinants of oil prices in today's rapidly changing global

* Corresponding author. Tel.: +0-000-000-0000 ; fax: +0-000-000-0000.
E-mail address: siham.ak2@gmail.com

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scenario [6]. The digital age, characterized by vast amounts of data and advancements in computational capabilities, offers new opportunities for forecasting [7].

This research leverages deep learning to enhance oil price forecasting accuracy [8]. By integrating both time series oil price data and key economic indicators into a multimodal neural network architecture, we aim to capture the complex interplay of factors influencing oil prices [9]. In the context of market dynamics in renewable energy, a deeper understanding of oil price dynamics can shed light on the challenges and strategies for the renewable energy transition [10].

The rest of the paper is structured as follows: Section 2 considers related work and establishes the current state of the art in oil price forecasting. In Section 3, we detail the methodology employed, including the data acquisition, pre-processing and architectural refinement of our multi-modal deep learning model. Results and their implications are presented in Section 4, followed by a comprehensive discussion in Section 5. We conclude the article in Section 6, also outlining potential avenues for future research.

2. Literature Review

The forecasting of oil prices has been a focal topic in energy economics and financial research for several decades. This section reviews the primary methods and findings in the literature, categorizing them into traditional econometric models, machine learning techniques, and recent advances in deep learning. Historically, the primary approach to predicting oil prices involved econometric models. Autoregressive integrated moving average (ARIMA) models have been popular in the oil price forecasting literature due to their capacity to model time series data [5]. However, they mainly capture linear patterns and might struggle with the nonlinear dynamics of oil prices [11]. Vector autoregressions (VAR) and structural VAR (SVAR) models have been employed to capture the interdependencies between oil prices and other macroeconomic variables [20]. They provide insights into the dynamic interactions among these variables but may miss more intricate underlying patterns. Recently, machine learning techniques have been introduced to the domain of oil price prediction. Approaches such as support vector machines (SVM) [21] and random forests [22] have shown promising results by effectively capturing non-linear relationships inherent in the data. Deep learning, particularly recurrent neural networks (RNNs) and long short-term memory (LSTM) networks, has gained traction in oil price forecasting [23]. LSTMs, with their ability to remember long-term dependencies in sequences, have shown potential in capturing the volatile nature of oil prices [24].Additionally, the introduction of multimodal deep learning, which combines different types of data sources into a unified model, presents a novel approach. By integrating time series data with economic indicators or textual news data, these models aim to offer a holistic prediction framework [12]. While deep learning models, especially LSTMs, have shown superior performance in many forecasting tasks, there's limited understanding of their applicability in multimodal scenarios, especially in the context of oil price forecasting. Most studies focus on single-source data, either time series or economic indicators, overlooking the potential synergies of combining them. Moreover, the interpretability of deep learning models remains a challenge, making it crucial to explore techniques that provide insights into model decisions

3. Methodology

Fig. 1 illustrates the comprehensive methodology employed in our study, providing a visual representation of the entire process, from data collection to model interpretation.

3.1. Data Collection

In this research, we exploit a comprehensive dataset that captures both the time-series fluctuations of West Texas Intermediate (WTI) crude oil prices and an assortment of important economic indicators. Spanning from April 17, 2013 to January 31, 2023, the dataset offers a robust perspective on complex oil market dynamics alongside global economic trends. The dataset offers insights into the closing prices of various indices such as the WTI Crude Oil Spot, GSCI Commodities Index, S&P 500, NASDAQ, and DJIA, among others. Additionally, it includes economic metrics like the Effective Federal Funds Rate, Economic Policy Uncertainty Index, and various treasury rates. A particular highlight of this dataset is the inclusion of monthly U.S. oil supply and demand figures, allowing for an exploration of

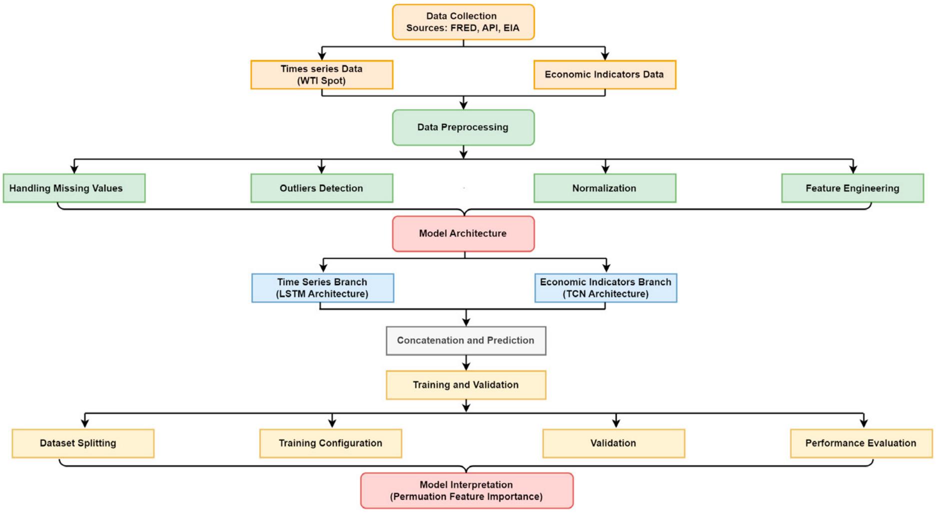

Fig. 1. Multimodal architecture combining oil price data and economic indicators for forecasting.

supply-demand dynamics in the context of oil price variations. The primary sources of this dataset include economic data, index prices, and rates, which were obtained from the Federal Reserve Economic Data (FRED) repository [13]. The US Oil Demand data was procured from the American Petroleum Institute (API) [14], while the US Oil Supply data was sourced from the U.S. Energy Information Administration (EIA) [15].

3.2. Data Preprocessing

Appropriate pre-processing of the dataset is essential to ensure the robustness and reliability of the derived Machine Learning (ML) models. This study implemented a systematic preprocessing procedure, comprising the following steps:

3.2.1. Handling Missing Values

Considering the long time period of the dataset, missing values have been anticipated occasionally. A preliminary exploration confirmed their presence in several columns. To maintain data integrity without introducing bias, linear interpolation, a common technique for time series data [16], was employed to impute these missing entries.

3.2.2. Detecting and handling outliers

Outliers can significantly distort the training of deep learning models, resulting in sub-optimal performance. The dataset was subjected to outlier analysis using the IQR (Interquartile Range) method, a widely used statistical approach [17]. Observations falling below the first quartile or above the third quartile were considered outliers. These outliers were capped at the respective thresholds, thus preserving the overall distribution of the data while minimizing extreme influences [18].

3.2.3. Normalization

The dataset comprises features with varying scales, such as oil prices, economic indicators, and index prices. Given that neural networks are sensitive to input scales [19], MinMaxScaler was employed to normalize the dataset, ensuring all feature values lie between 0 and 1.

3.2.4. Feature Engineering

In time-series forecasting, the term lag refers to the difference in time between two periods. For our model, we introduce lag features to exploit past data in our forecasts. Specifically, we use an offset of 3, which means that for each data point at time t, we also supply the model with data from $t-1$, $t-2$ and $t-3$. This method equips the model with a short-term memory of the last three time steps when it comes to forecasting future prices. In doing so, the model can recognize short-term patterns and trends in the data that may not be immediately obvious from the current data point alone.

3.3. Model Architecture

In this research, a multimodal deep learning model was designed to exploit both time series data related to WTI oil prices and structured economic indicators. The model architecture is a combination of RNN layers, in particular LSTM layers, for sequential time series data and TCN layers for economic indicators. The architecture can be described as follows:

3.3.1. Time Series Branch

The multimodal architecture comprises two main branches. The time series branch begins with an input layer specifically designed for time series sequences, followed by two LSTM layers capable of capturing the temporal subtleties of the data. In parallel, the economic indicator data path starts with an input layer designed for structured economic indicators. This path uses a sequence of TCN layers to discern temporal patterns, followed by a dense layer to refine the results. Finally, the two branches converge in the concatenation and prediction stage. Here, their outputs are merged, processed by two dense layers and routed to a single output neuron responsible for predicting the price of WTI oil. The detailed parameters of each layer are given in Table 1.

Layer Type	Parameters	Description
Input (Time Series Data)	-	Shape: (3, 1), 3 lag features for time series
LSTM Layer	200	50 neurons, return_sequences=True
LSTM Layer (after flattening)	50	Flattened output from the LSTM layer
Input (Economic Indicators)	-	Shape: (3, 14), 3 lag features for each of the 14 economic indicators
TCN	Varied	Multiple layers with dilated convolutions
TCN Output (after flattening)	128	Flattened output from the TCN layers
Concatenate	-	Combining LSTM and TCN outputs
Dense Layer	94	94 neurons, relu activation
Dense Layer (Output)	1	1 neuron, linear activation

Table 1. Parameters of the multimodal deep learning model

3.4. Training and Validation

3.4.1. Dataset Splitting

To evaluate the model's performance effectively, the dataset was split into training, validation, and test subsets. The training set, constituting 70% of the data, was used for model training. The validation set, comprising 15%, assisted in hyperparameter tuning and early stopping, while the test set, also 15%, was reserved for the final evaluation of the model's performance.

3.4.2. Training Configuration

The model's training utilized the Adam optimizer with a learning rate of $1x10^{-3}$. Given the regression nature of our task, the Mean Squared Error (MSE) was chosen as the loss function. The model was trained over 100 epochs with a batch size of 32. To ensure the model's robustness and to reduce overfitting, dropout layers with a dropout rate of 0.2 were integrated into the architecture.

3.4.3. Validation

Throughout the training process, the model's performance was continuously monitored on the validation set. This ongoing assessment facilitated the early detection of overfitting, callbacks such as Early Stopping were employed.

This technique interrupted the training process when the validation loss no longer showed improvement, ensuring the model did not become overly adapted to the training data.

3.4.4. Performance Metrics

The model's proficiency on the test set was measured using multiple metrics to provide a comprehensive evaluation:

- Mean Absolute Error (MAE): Represents the average of the absolute differences between the predicted and actual values [25]
- Mean Squared Error (MSE): Provides the average squared differences between the predicted and actual values [26].
- Root Mean Squared Error (RMSE): Represents the square root of the MSE, giving a sense of how much, on average, the model's predictions deviate from the actual values [27].
- Directional stat (Dstat): Measures the percentage of predictions that correctly predict the direction of price movement [28].

3.5. Model Interpretation

Deep learning models, particularly complex architectures like the one employed in this study, are often criticized for their lack of transparency, commonly referred to as black boxes models. Understanding how these models make decisions is crucial, especially in sectors like finance and economics where the challenges are high. Model interpretation not only increases trust in model predictions but also provides insights that can guide decision-making processes [29, 30]. To understand which input features are most influential in predicting the oil prices, we employed an interpretation method known as Permutation Feature Importance (PFI) [31]. PFI offers a technique to measure the importance of each feature by observing the decrease in the model's performance when the feature's value is randomly shuffled. This process breaks the relationship between the feature and the target, thus any drop in performance is indicative of the feature's importance. For our multimodal model, feature importance was calculated separately for time series data and economic indicators. This separation provided insights into which historical time series data points and which economic indicators played significant roles in the forecasting of the oil price. The TCN part of our model focuses on the economic indicators. By interpreting this component, we were able to identify patterns and trends over time. This temporal insight is essential as it reveals how different economic indicators influence oil prices over different time horizons.

4. Results and Discussion

In this section, we evaluate the effectiveness of our multimodal deep learning approach in forecasting oil prices and analyze the importance of the underlying economic indicators.

4.1. Forecasting Performance

The primary objective of this study was to develop a model capable of accurately forecasting oil prices using a multimodal deep learning approach. To visualize the model's performance, we plotted the actual versus predicted values of the WTI Crude Oil Spot prices, as shown in Fig. 2. Examining this Figure , several observations can be made:

- Closeness of Predicted to Actual Values: The model's predicted values (orange line) closely follow the actual prices (blue line), showcasing the model's ability to capture the underlying patterns in the data.
- Capturing Volatility: Oil prices are known for their volatility, and the model seems adept at capturing sudden price changes, further emphasizing its robustness.
- Directional Accuracy: The directional changes in the predicted values mimic those in the actual prices, corroborating the high Dstat value of 97.79%.

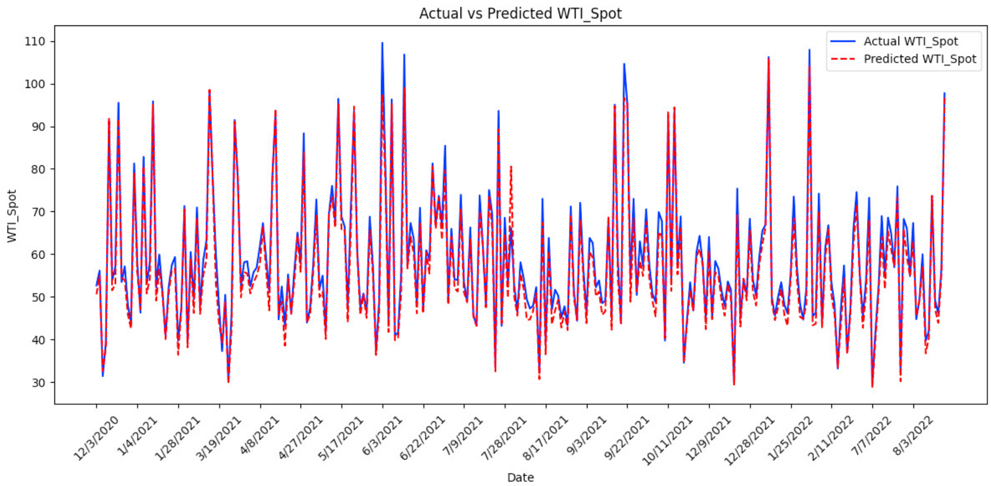

Fig. 2. Actual vs. Predicted WTI Crude Oil Spot Prices.

In terms of numerical metrics:

- The MAE value of 0.0262 indicates that the model's forecasts were, on average, off by an absolute value of 0.0262 units from the actual oil prices. Such precision, given the challenges associated with oil price forecasting, is commendable.
- The MSE and RMSE values, 0.00109 and 0.0330 respectively, underline the model's consistency in predictions.
- A Dstat of 97.79% confirms the model's efficacy in predicting the correct direction of oil price movements .

4.2. Economic Indicators' Influence

For a comprehensive overview of oil price forecasts, it is essential to understand the influence of the various economic indicators. Economic indicators serve as indicators of fundamental economic activity and conditions, and their dynamics play a decisive role in determining future oil price trends. Table 2 shows the ranking of economic indicators based on their importance scores derived from PFI. This method provides a numerical measure indicating the weight of each characteristic in the prediction model. The three main influential indicators are:

- GSCI (Commodities Index): This index has the most significant influence on oil prices in our study. Commodity indices such as the GSCI provide a broad measure of the performance of the commodity sector, and crude oil is an important component of these indices. A strong correlation between the commodity index and oil prices indicates that general commodity market dynamics have a significant impact on oil prices.
- Economic uncertainty index: Economic policy uncertainty can lead to volatility in oil prices, as businesses and consumers may delay spending and investment, affecting demand. In addition, geopolitical tensions and supply disruptions can lead to uncertainty in the oil market, making this indicator essential.
- NASDAQ: Stock market indices such as the NASDAQ can serve as barometers for overall economic health and investor sentiment. Positive sentiment on stock markets can be correlated with increased economic activity, leading to higher demand for oil.

Among the least influential indicators, the effective Federal Funds Rate and the 5-year break-even inflation rate had minimal influence. Although these indicators are of macroeconomic importance, their direct impact on oil price fluctuations could be reduced compared to other, more dominant factors. Interestingly, the least influential characteristic, according to our model, is the difference between 10-year and 2-year Treasury rates (10Y_Less_2Y). While this is an essential measure for understanding the yield curve and potential recession signals, its direct influence on

Rank	Economic Indicator	Importance Score
1	GSCI (Commodities Index)	0.028326169833257326
2	Economic Uncertainty Index	0.004556822166869467
3	NASDAQ	0.0019655201568735976
4	US 3-month treasury	0.0016651283273923297
5	DJIA	0.0012120984226836556
6	US 1-year treasury	0.0011097281554482833
7	Monthly US Oil Demand	0.0011064771665022556
8	S&P 500	0.0008895393546928135
9	WTI Crude Oil Spot	0.0007853579404041794
10	US 10-year minus 2-year treasury	0.0007433123755675691
11	5-year break even inflation rate	0.0005130632229451248
12	DXY	0.00030295431583807013
13	Effective Federal Funds Rate	2.144060385313542e-05

Table 2. Permutation Feature Importance for Economic Indicators

oil prices in the context of our model is minimal. However, the clear ranking of economic indicators underlines the complex interplay of different macroeconomic factors in determining oil prices. While some indicators have a direct and pronounced impact, others play a more discrete role, highlighting the complexity of the global oil market. To understand the dynamics of oil prices is not only crucial for the oil industry, but also has significant implications for the renewable energy market. Here are the main aspects of our study and their implications:

- **Correlation between Oil Prices and Economic Indicators**: Economic indicators such as the GSCI, the Economic Uncertainty Index and the NASDAQ influence oil prices considerably, reflecting the close link between the global economy and the oil market. High oil prices stimulate interest in renewable energies, but low oil prices mean that renewable energies face tougher competition.
- **The impact of uncertainty on renewable energies** : Our model reveals the role of the economic uncertainty index in oil prices. High uncertainty can inhibit investment in renewable energies. Therefore, stable economic conditions with low uncertainty are essential for the growth of the renewable energy sector.
- **Oil Prices and Renewable Transition**: Our research, focusing on oil price forecasts, highlights the challenges posed by renewable energies. Low oil prices could reduce the attractiveness of renewables. Renewable energy strategists therefore need to factor oil price fluctuations into their planning.

5. Conclusion and Future Works

Our multimodal deep learning model efficiently forecasts oil prices by integrating time series data and economic indicators. The results highlight the deep interaction between global economic factors, oil prices and the outlook for renewable energies. Further research can integrate more diverse data sources for greater accuracy. In addition, exploring advanced neural architectures and assessing the impact of climate-related policies on oil and renewable energy markets would offer broader perspectives. This study represents a fundamental step towards understanding the nuances of the energy economy in a sustainable future.

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